

THETFORD MINES, QC, Canada

WMO: 714940

Lat: 46.0492N

Lon: 71.2661W

Elev: 430

StdP: 96.26

Time Zone: -5.00 (NAE)

Period: 06-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -25.9	(c) -23.4	(d) -29.7	(e) 0.3	(f) -25.2	(g) -27.1	(h) 0.3	(i) -22.6	(j) 10.2	(k) -5.5	(l) 9.0	(m) -7.3	(n) 3.1	(o) 270	(p) 0.604

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.8	27.1	19.6	25.6	18.7	24.2	18.1	21.2	24.8	20.2	23.5	19.4	22.6	4.6	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
20.1	15.6	22.8	19.2	14.7	21.9	18.2	13.8	21.2	63.7	25.1	60.3	23.5	57.3	22.4	24.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 8.4	(b) 7.5	(c) 6.7	(e) -29.7	(f) 29.7	(g) 2.4	(h) 0.9	(i) -31.4	(j) 30.3	(k) -32.9	(l) 30.9	(m) -34.2	(n) 31.4	(o) -36.0	(p) 32.0		
(a) 8.4	(b) 7.5	(c) 6.7	(e) -29.9	(f) 23.0	(g) 2.4	(h) 0.8	(i) -31.6	(j) 23.6	(k) -33.0	(l) 24.0	(m) -34.3	(n) 24.4	(o) -36.1	(p) 25.0		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	4.2	-11.2	-10.3	-5.0	2.9	10.8	15.1	18.1	17.0	13.2	6.4	-0.6	-7.5
	DBStd	11.61	7.28	6.28	6.34	5.10	4.55	3.51	2.96	2.89	4.22	4.69	5.48	6.51
	HDD10.0	2966	658	567	465	221	45	3	0	0	16	132	318	542
	HDD18.3	5242	917	801	722	463	236	107	40	59	160	371	567	800
	CDD10.0	841	0	0	1	7	71	157	252	218	113	20	1	0
	CDD18.3	75	0	0	0	0	3	11	34	19	7	0	0	0
	CDH23.3	498	0	0	0	4	43	89	216	107	39	0	0	0
	CDH26.7	58	0	0	0	0	4	8	34	8	3	0	0	0
Wind	WSAvg	3.3	3.6	3.7	3.7	3.7	3.3	2.9	2.7	2.6	2.7	3.2	3.3	3.4
Precipitation	PrecAvg	1227	90	73	79	88	94	127	137	120	98	114	93	106
	PrecMax	1462	133	115	125	181	154	222	215	204	196	227	136	222
	PrecMin	887	39	37	28	13	48	44	75	29	46	41	40	50
	PrecStd	146	28	20	26	44	29	44	35	37	34	47	25	42
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	8.3	8.4	14.3	22.2	27.3	27.9	29.6	27.7	26.7	21.0	15.7	8.3
		MCWB	7.7	7.4	11.4	14.2	17.4	19.2	21.3	20.8	20.7	15.9	12.1	7.4
	2%	DB	5.2	3.8	8.6	17.5	24.3	25.8	27.4	26.0	24.3	18.4	12.2	5.2
		MCWB	4.1	2.6	4.9	10.6	15.9	18.1	20.2	19.4	18.8	14.0	9.7	4.6
	5%	DB	1.8	1.2	5.9	13.8	21.6	24.0	25.7	24.4	22.4	16.0	9.5	3.0
		MCWB	1.2	-0.1	2.7	8.4	14.2	17.4	19.2	18.5	17.7	12.6	7.4	2.4
	10%	DB	-0.8	-0.6	3.8	11.0	19.0	21.9	24.1	22.9	20.5	13.6	7.3	0.9
		MCWB	-1.4	-1.4	1.4	6.6	13.0	16.1	18.2	17.8	16.8	11.1	5.7	0.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	7.8	7.7	11.2	15.4	20.0	21.2	23.0	22.0	21.5	17.1	13.1	7.5
		MCDB	8.2	8.4	14.4	21.2	24.0	25.1	28.0	26.3	24.8	18.3	14.1	8.2
	2%	WB	4.8	2.8	6.1	12.1	17.5	20.1	21.6	20.6	20.1	15.0	10.4	4.5
		MCDB	5.0	3.6	7.4	16.1	21.8	23.5	25.2	23.8	23.0	17.3	12.4	4.9
	5%	WB	1.0	0.2	3.3	9.1	15.8	18.7	20.4	19.7	18.8	13.4	8.0	2.4
		MCDB	1.8	0.8	5.1	12.3	19.4	22.1	23.7	22.6	21.3	15.6	9.1	3.0
	10%	WB	-1.4	-1.4	1.6	7.0	14.1	17.3	19.5	18.8	17.5	11.5	5.8	0.2
		MCDB	-0.8	-0.6	3.6	10.5	17.5	20.4	22.8	21.8	19.8	13.3	7.3	0.6
Mean Daily Temperature Range	5% DB	MDBR	9.1	9.3	9.4	9.3	10.9	10.0	9.8	9.6	10.0	8.1	7.3	7.5
		MCDBR	9.6	10.8	11.6	14.5	14.4	12.9	11.5	11.4	11.6	11.4	10.9	8.9
		MCWBR	9.6	9.9	8.7	9.2	7.8	6.5	5.5	5.8	6.6	7.8	8.7	9.2
	5% WB	MCDBR	9.6	10.1	9.8	12.2	11.4	9.8	9.6	9.3	10.2	10.1	9.8	9.0
MCWBR		9.8	9.7	8.4	8.8	7.6	6.2	5.4	5.4	6.8	8.0	8.9	9.2	
Clear-Sky Solar Irradiance	taub	0.254	0.269	0.303	0.327	0.350	0.388	0.391	0.376	0.347	0.317	0.291	0.276	
	taud	2.370	2.293	2.322	2.430	2.428	2.305	2.309	2.356	2.437	2.524	2.513	2.355	
	Ebn at Noon	864	914	927	932	917	878	870	870	869	853	816	785	
	Edh at Noon	71	95	108	107	111	126	124	114	96	76	64	66	
All-Sky Solar Radiation	RadAvg	1.40	2.29	3.51	4.40	4.96	5.49	5.43	4.76	3.81	2.21	1.31	0.95	
	RadStd	0.20	0.29	0.28	0.42	0.54	0.37	0.50	0.30	0.33	0.20	0.17	0.10	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (41 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page